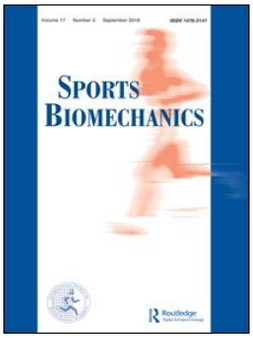

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Poor hamstrings-to-quadriceps torque ratios in male soccer players: weak hamstrings, strong quadriceps, or both?

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ABSTRACT

This study aimed to evaluate the influence of hamstrings and quadriceps strength on the hamstrings-to-quadriceps conventional ($H:Q_{con}$) and functional ($H:Q_{fun}$) ratios in male soccer players. Quadriceps concentric peak torque (PT) and hamstrings concentric and eccentric PT were assessed with isokinetic dynamometry at 60°/s in 101 players (202 legs). The cut-points of 0.50, 0.55 and 0.60 were used to assess muscle imbalance through the $H:Q_{con}$ ratio, while 0.80, 0.85 and 0.90 were used for $H:Q_{fun}$ ratio. Legs with lower $H:Q_{con}$ ratio had decreased hamstrings concentric PT ($p < 0.01$; moderate and large effect sizes) and increased quadriceps concentric PT ($p < 0.01$; moderate effect sizes) in all cut-points. Legs with lower $H:Q_{fun}$ ratio had decreased hamstrings eccentric PT ($p < 0.01$; large effect sizes) for all cut-points, and controversial results for quadriceps concentric PT ($p < 0.01$ only for 0.80 cut-point; small effect sizes). $H:Q_{con}$ ratio presented only weak correlations with quadriceps ($r = -0.37$) and hamstrings ($r = 0.45$) concentric PT, while $H:Q_{fun}$ ratio had a negligible correlation with quadriceps concentric PT ($r = -0.30$) and a moderate correlation with hamstrings eccentric PT ($r = 0.66$). In conclusion, our findings support that hamstrings strength deficit is the key factor for low H:Q ratios in male soccer players, especially those with poor $H:Q_{fun}$ ratio.

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KEYWORDS
Football; strength ratio; knee extensor; knee flexor; muscle imbalance

Introduction

High injury rate has been a serious concern in professional male soccer for many years (Hawkins & Fuller, 1998). Data from European elite teams support an average of 8 injuries per 1.000 hours of exposure (4.1/1000 h in training and 27.5/1000 h in matches), and each player usually misses 37 days due to injuries each season (i.e. ~12% of the entire season) (Ekstrand et al., 2011). Hamstrings strain injury (HSI) is the most prevalent non-contact injury (12% of all injuries) (Ekstrand et al., 2011). A team with a 25 player-squad typically has 5–6 players with at least one HSI per season, with an average layoff time of 17 days per injury (Ekstrand et al., 2016). In addition, 13% of players sustain a recurrent HSI within two months after return to play (Ekstrand et al., 2016) and the rehabilitation

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of re-injured players is usually more time-consuming (Heiderscheit et al., 2010). Therefore, the 'injury burden' (i.e. the cross-product of severity and incidence) of HSI is one of the heaviest in soccer (Bahr et al., 2017).

HSI is typically a non-contact injury, and 60–80% of the cases in soccer occur during high-speed running (Timmins et al., 2016; Woods et al., 2004). Consequently, intrinsic risk factors seem to play a key role in HSI. Among a range of modifiable risk factors, clinicians working in premier league teams around the world consider 'muscle imbalance' within the top-3 risk factors for non-contact injuries in professional soccer (McCall et al., 2014; Meurer et al., 2017). Although 'muscle imbalance' is a broad expression, the strength imbalance between knee flexor and extensor muscle groups has been the focus of studies interested in HSI. Classically, the highest isokinetic torque value generated by knee flexor muscles along the range of motion (often called 'hamstrings peak torque'—PT) is divided by the highest torque generated by knee extensor muscles (often called 'quadriceps PT') to estimate the agonist-antagonist torque ratio known as the 'hamstrings-to-quadriceps (H:Q) ratio' (Baroni et al., 2020). The 'H:Q conventional ratio' ($H:Q_{con}$; hamstrings concentric PT divided by quadriceps concentric PT) is the oldest and the most described strength ratio (Coombs & Garbutt, 2002; Dvir et al., 1989), while the 'H:Q functional ratio' ($H:Q_{fun}$; hamstrings eccentric PT divided by quadriceps concentric PT) is believed to better represent the muscle requirements of sports (e.g., running and kicking) (Coombs & Garbutt, 2002; Dvir et al., 1989).

Even in face of the lack of consensus regarding the predictive value of H:Q ratios for sport injuries (e.g., see Dauty et al., 2018; Van Dyk et al., 2017), isokinetic testing is largely used by premier league soccer teams as part of preseason screening tests (McCall et al., 2014; Meurer et al., 2017). Reference values of 0.5–0.6 (Croisier et al., 2002, 2008; Lee et al., 2018; Yeung et al., 2009) and 0.8–1.0 (Croisier et al., 2002, 2008; Lee et al., 2018) have been used to assess muscle imbalance through $H:Q_{con}$ and $H:Q_{fun}$ ratios, respectively. A recent systematic review including 30 articles (1,727 male professional soccer players) reported that coaching staffs should expect $H:Q_{con}$ ratio values around 0.6 and $H:Q_{fun}$ ratio values around 0.8 when tests are performed at 60°/s (the most used isokinetic angular velocity) (Baroni et al., 2020). Generally, a player is considered with a proper strength balance when his H:Q ratio is equal to or greater than the cut-offs recommended by literature. However, the H:Q ratio can be compromised by hamstrings weakness, by quadriceps excessive strength, or both.

Previous studies have shown diverse findings regarding the cause of muscle imbalance in soccer players. While one study showed that male soccer players with low $H:Q_{con}$ ratio have weaker hamstrings and stronger quadriceps than those with high $H:Q_{con}$ ratio (Greco et al., 2012), another reported only lower hamstrings PT in players with low $H:Q_{con}$ ratio compared to those with high $H:Q_{con}$ ratio, with no between-group difference for quadriceps PT (De Abreu Camarda & Denadai, 2012). A third study found a third result and showed that players with low $H:Q_{con}$ ratio had only greater quadriceps PT compared to those with high $H:Q_{con}$ ratio (Bogdanis & Kalapotharakos, 2016). These conflicting results did not make clear the causes of low H:Q ratios in soccer players: weak hamstrings, strong quadriceps, or both. Thus, the aim of this study was to verify the influence of hamstrings and quadriceps PTs on the $H:Q_{con}$ and $H:Q_{fun}$ ratios in male soccer players. Our hypothesis was that athletes with

muscle imbalance would have weaker hamstrings in comparison with athletes without muscle imbalance.

Methods

Experimental approach to the problem

A cross-sectional study was conducted at the Physiotherapy Laboratory of the *Federal University of Health Sciences of Porto Alegre* (Porto Alegre, Brazil). Cut-points for H:Q ratios were defined, and the quadriceps and hamstrings PTs were compared between players above and below that cut-points. In addition, correlation analyses between PTs and H:Q ratios were performed.

Participants

The study protocol was approved by the *Federal University of Health Sciences of Porto Alegre* Ethics Committee. All volunteers signed a written informed consent before their participation.

Two soccer teams competing in one of the major state divisions in Brazil agreed to participate in this study. One-hundred and one male soccer players were assessed (52 professionals and 49 under-20 players). Age (20.0 ± 3.4 years), height (179.9 ± 7.6 cm), and body mass (74.8 ± 8.5 kg) were measured before testing. Twelve players were goalkeepers, 24 were defenders, 41 were midfielders, and 24 were forwards. Professional and under-20 players followed a similar training routine, which usually consisted of two daily training sessions, three to five days per week according to official matches and trips on each week. Players who were not participating in the full team training routine due to injuries or any other reasons were not considered for analysis in the study. Players with history of HSI or other injuries in adjacent body sites were included since they were considered recovered by the medical departments and were engaged in the full training routine.

Procedures

All assessments were performed during the two first weeks of the pre-season period. The coaching staff was informed that players should not perform vigorous training sessions in the day before testing. In addition, athletes received the following recommendations: (1) not to perform high-intensity physical activities in the 24 hours prior to the tests; (2) not to take any kind of analgesic and/or anti-inflammatory drugs within 48 hours prior to testing; (3) not to consume stimulant substances (e.g., caffeine) at day of the tests.

After a general warm-up (5 minutes of cycle-ergometer exercise), athletes were positioned on the isokinetic dynamometer (Biodex System 4; Biodex Medical Systems, USA) according to the manufacturer recommendations and followed the testing protocol used routinely in our lab (Ribeiro-Alvares et al., 2019). Athletes were in the seating position with 80° of hip flexion and 90° of knee flexion (0° = full knee extension) and stabilised with shoulder, waist and thigh straps. Athletes performed 10 concentric knee flexion/extension repetitions at 90°/s with a submaximal effort level for specific warm-up and familiarisation with the equipment. Thereafter, two attempts of three consecutive maximum contractions were executed in the concentric-concentric mode (60°/s; 0–90° of

knee flexion), i.e. reciprocal quadriceps concentric contraction followed by hamstrings concentric contraction. Then, two attempts of three consecutive maximal contractions in the eccentric-eccentric mode (60°/s; 30–90° of knee flexion), i.e. reciprocal quadriceps eccentric contraction followed by hamstrings eccentric contraction. A 1-min resting period was respected between attempts. Both legs were tested in a randomised order between participants. Quadriceps concentric PT and hamstrings concentric and eccentric PTs were used for data analysis, as well as to estimate muscle imbalance. $H:Q_{con}$ ratio (hamstrings concentric PT/quadriceps concentric PT) and $H:Q_{fun}$ ratio (hamstrings eccentric PT/quadriceps concentric PT) were calculated (Baroni et al., 2020).

Statistical analyses

Mean and standard deviations (SD) were calculated for quadriceps concentric PT, hamstrings concentric and eccentric PTs, as well as for $H:Q_{con}$ and $H:Q_{fun}$ ratios for all legs tested. Different cut-points for H:Q ratios were used for analysis. The cut-points of 0.50, 0.55 and 0.60 were assumed for the $H:Q_{con}$ ratio; while the cut-points of 0.80, 0.85 and 0.90 were used for the $H:Q_{fun}$ ratio. The classical cut-point of 1.0 for $H:Q_{fun}$ ratio was not used due to the reduced number of legs with values above this cut-point (n = 6). Pairwise comparisons of quadriceps and hamstrings PTs of each group (i.e. below and above the cut-points for each ratio) were made using paired t-test, assuming a significance level of $p < 0.01$. Effect sizes were calculated using Cohen's d, assuming the results as a trivial ($d < 0.2$), small ($d < 0.5$), moderate ($d < 0.8$) or large ($d > 0.8$) (Cohen, 1977). Only moderate or greater effect sizes were considered clinically relevant. Pearson correlation coefficient (r) was applied to assess the following relationships: quadriceps concentric PT with $H:Q_{con}$ and $H:Q_{fun}$ ratios; hamstrings concentric PT with $H:Q_{con}$ ratio; and hamstrings eccentric PT with $H:Q_{fun}$ ratio. Correlations were classified as negligible ($r < 0.3$), weak ($r = 0.3–0.5$), moderate ($r = 0.5–0.7$), strong ($r = 0.7–0.9$) and very strong ($r > 0.9$) (De Araujo Ribeiro Alvares et al., 2015).

Results

From all the 202 legs tested, quadriceps concentric PT, hamstrings concentric PT, hamstrings eccentric PT, $H:Q_{con}$ ratio and $H:Q_{fun}$ ratio were 273.5 ± 40.7 Nm, 150.2 ± 22.6 Nm, 208.5 ± 39.2 Nm, 0.5 ± 0.1 and 0.8 ± 0.1, respectively.

Table 1. Quadriceps (Q) and hamstrings (H) concentric peak torques (PT) (mean ± SD) of legs with hamstrings-to-quadriceps conventional ($H:Q_{con}$) ratio below (unbalanced) and above (balanced) the cut-points.

$H:Q_{con}$ cut-points	Unbalanced	Balanced	p-value	Effect size [95%CI]	Δ%*
0.50	n = 52	n = 150			
- Q concentric PT	290.35 ± 47.88	267.68 ± 36.37	0.003	0.58 [−12.30–6.44]	8.47%
- H concentric PT	137.05 ± 20.82	154.77 ± 21.57	<0.001	0.83 [−2.62–6.49]	−11.45%
0.55	n = 98	n = 104			
- Q concentric PT	286.07 ± 42.88	261.69 ± 34.90	<0.001	0.63 [−7.86–7.34]	9.32%
- H concentric PT	142.24 ± 20.61	157.71 ± 22.09	<0.001	0.73 [−3.52–4.81]	−9.81%
0.60	n = 153	n = 49			
- Q concentric PT	278.95 ± 40.50	256.55 ± 37.04	<0.001	0.57 [−5.85–10.94]	8.73%
- H concentric PT	145.40 ± 20.49	165.20 ± 22.91	<0.001	0.94 [−5.47–4.19]	−11.98%

* Percentage change compared with the balanced limb.

Table 2. Quadriceps (Q) and hamstrings (H) eccentric peak torques (PT) (mean ± SD) of legs with hamstrings-to-quadriceps functional ($H:Q_{fun}$) ration below (unbalanced) and above (balanced) the cut-points.

$H:Q_{fun}$ cut-points	Unbalanced	Balanced	p-value	Effect size [95% CI]	Δ%*
0.80	n = 118	n = 84			
- Q concentric PT	279.65 ± 42.16	264.90 ± 37.26	0.009	0.37 [−7.24–8.34]	5.77%
- H eccentric PT	189.92 ± 30.31	234.69 ± 35.02	<0.001	1.39 [−6.10–6.86]	−19.08%
0.85	n = 145	n = 57			
- Q concentric PT	277.64 ± 41.41	263.04 ± 37.38	0.017	0.36 [−6.38–10.07]	5.55%
- H eccentric PT	195.85 ± 32.79	240.82 ± 35.61	<0.001	1.34 [−7.90–6.68]	−18.67%
0.90	n = 174	n = 49			
- Q concentric PT	276.44 ± 40.11	256.55 ± 37.04	0.522	0.51 [−5.45–10.88]	7.75%
- H eccentric PT	202.36 ± 35.31	246.94 ± 40.82	<0.001	1.23 [−10.21–6.46]	−18.05%

*Percentage change compared with the balanced limb.

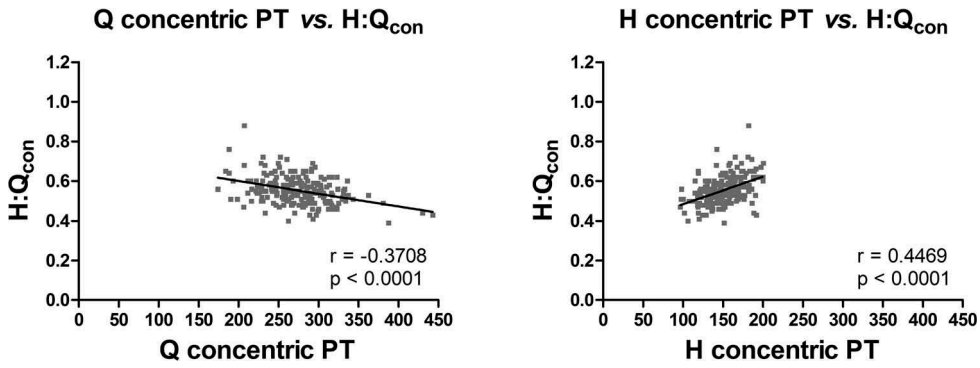

Figure 1. Correlations between hamstrings (H) and quadriceps (Q) peak torques (PT) (Nm) and $H:Q_{con}$.

Independently of the cut-point chosen for $H:Q_{con}$ ratio, legs with lower ratio values had decreased hamstrings concentric PT ($p < 0.01$; moderate and large effect sizes) and increased quadriceps concentric PT ($p < 0.01$; moderate effect sizes) compared to legs with greater values of $H:Q_{con}$ ratio (Table 1).

For the three cut-points chosen for $H:Q_{fun}$ ratio, legs with lower ratio values had decreased hamstrings eccentric PT ($p < 0.01$; large effect sizes) in comparison with legs with lower values of $H:Q_{fun}$ ratio (Table 2). Legs with lower values of $H:Q_{fun}$ ratio had increased quadriceps concentric PT only for the cut-point of 0.80 ($p < 0.01$; small effect size; Table 2).

Both hamstrings and quadriceps concentric PTs presented weak positive and negative correlations with the $H:Q_{con}$ ratio values, respectively (Figure 1). In contrast, while quadriceps concentric PT had negligible correlation with the $H:Q_{fun}$ ratio, hamstrings eccentric PT showed a moderate positive correlation with the $H:Q_{fun}$ ratio (Figure 2).

Discussion and implications

This study aimed at elucidating the influence of hamstrings and quadriceps strength on the H:Q ratios in male soccer players. In short, our findings support that low $H:Q_{con}$ ratio values in soccer players are caused by both strengthened quadriceps and weakened

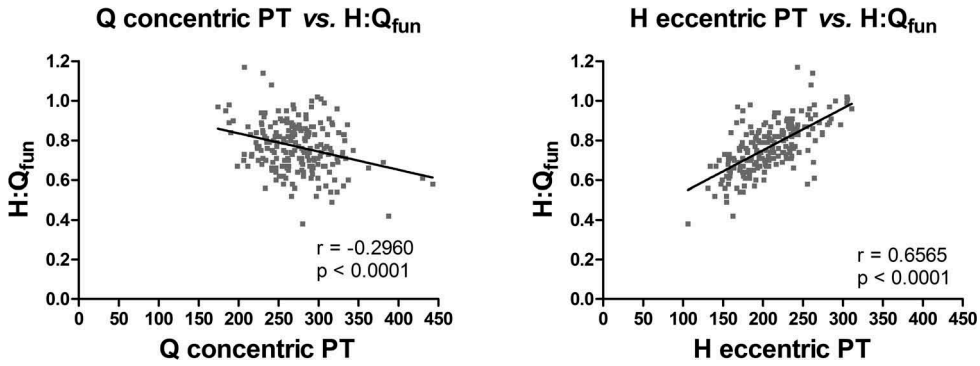

Figure 2. Correlations between hamstrings (H) and quadriceps (Q) peak torques (PT) (Nm) and $H:Q_{fun}$.

hamstrings, while low $H:Q_{fun}$ ratio values should be attributed to decreased hamstrings eccentric strength.

Since the introduction of isokinetic dynamometers in the 1960's (Hislop & Perrine, 1967), investigations have been made on the hamstrings-to-quadriceps strength balance in several athletic populations (Croisier et al., 2002, 2008; Dauty et al., 2018; Orchard et al., 1997; Van Dyk et al., 2016; Yeung et al., 2009). Reports were concerned only with the $H:Q_{con}$ ratio until the late 1980's, when studies started to assess the so called 'dynamic control ratio' and now known as the $H:Q_{fun}$ ratio (Dvir et al., 1989). More recently, researchers have been trying to understand if the 'muscle imbalance' attributed to soccer players with low H:Q ratio values is primarily related to weakness of hamstrings muscles or to excessive strength of quadriceps muscles (De Abreu Camarda & Denadai, 2012; Bogdanis & Kalapotharakos, 2016; Greco et al., 2012). These previous studies included a limited number of players (18–39 participants) and presented conflicting results; thus, our study assessed a large sample size (101 players; 202 legs) to verify the weight of hamstrings and quadriceps strength for low H:Q ratio values in male soccer players.

When considering the results of the $H:Q_{con}$ ratio, the findings of the present study contrast to those from Bogdanis and Kalapotharakos (2016), who reported that unbalanced soccer players had quadriceps 16% stronger than the balanced ones, without between-group difference for hamstrings strength. Our findings are also different than those from De Abreu Camarda and Denadai (2012), who showed similar quadriceps PT values in unbalanced compared to balanced players, and hamstrings 17% weaker in the unbalanced group. However, in agreement to our findings, Greco et al. (2012) reported that unbalanced players had higher quadriceps (8%) and lower hamstrings (16%) PT compared to balanced players. Regarding the $H:Q_{fun}$ ratio, our results differ from the ones reported by De Abreu Camarda and Denadai (2012), who found that balanced soccer players had hamstrings eccentric PT 19% higher than unbalanced players, but no between-group difference for quadriceps PT. Further comparisons with the other two studies involving soccer players were not possible because the $H:Q_{fun}$ ratio was not assessed by Bogdanis and Kalapotharakos (2016) or Greco et al. (2012). Differences between our findings and those provided by previous studies may be related to the small number of players assessed in those investigations and to the specific cut-point values used to diagnose muscle imbalance in each study.

The present study showed that 76% of the assessed legs were below the traditional cut-point of 0.6 for the H:Q_{con} ratio (Coombs & Garbutt, 2002). In relation to the H:Q_{fun} ratio, 86% of the legs were below the cut-point of 0.9, and only 6% of legs presented a proper H:Q_{fun} ratio when considering the traditional cut-point of 1.0 (Coombs & Garbutt, 2002). The high prevalence of players with low H:Q_{con} ratio is surprising since a recent systematic review concluded that most professional male soccer players reach the landmark of 0.6 when tests are performed at 60°/s (Baroni et al., 2020). On the other hand, the high prevalence of players with low H:Q_{fun} ratio further support the systematic review's conclusion that the usual landmark of 1.0 is rarely reached by professional male soccer players when tested at slow angular velocities (Baroni et al., 2020). Therefore, two non-excluding hypotheses arise: (i) the soccer specific training leads to a suitable strength imbalance between hamstrings and quadriceps muscles; and/or (ii) the H:Q_{fun} ratio cut-point of 1.0 is inadequate.

The uncertainties about the most appropriate cut-point for H:Q ratios led us to analyse three cut-points for each H:Q ratio. In addition to the results provided by the null hypothesis tests (i.e. p-values), the effect sizes make clear that hamstrings weakness is the main responsible for low values in the H:Q_{con} ratio (see Table 1), and those are even more impactful to the low values in the H:Q_{fun} ratio (see Table 2). In a practical standpoint, our findings suggest that hamstrings strengthening is the key factor for soccer players with deficits on H:Q torque ratios, especially those with poor H:Q_{fun} ratio. Although both H:Q ratios may be also improved by a reduction in quadriceps strength, it does not make sense when the goal is injury prevention. Moreover, reduced quadriceps strength could impairs the athlete's performance (Hamilton et al., 2008), which is unacceptable in the context of professional sports. Therefore, players with poor H:Q ratios should be engaged in specific training programmes until reaching the proper level of muscle strength balance established by the coaching staff for their player-squad.

According the prospective study by Croisier et al. (2008) with a large sample of professional soccer players, the HSI rate was 4.66 times higher in players with untreated strength imbalances in comparison with those showing no imbalance in preseason, and 2.89 times higher for players with strength imbalances who had subsequent compensating training without subsequent isokinetic assessments; but normalising the isokinetic parameters reduced the reduced the injury rate to that observed in players without imbalances (relative risk = 1.43). Croisier et al. (2008) only described the intervention as a hamstrings conditioning programme under the team physical trainer or physical therapist supervision. Fortunately, a range of training programmes have been shown to be effective for increasing the hamstrings eccentric strength using isokinetic dynamometers (Ruas et al., 2018), strength training devices (Monajati et al., 2018), and bodyweight exercises (Monajati et al., 2016). Well-conditioned soccer players usually increase ~11% their hamstrings eccentric strength with 8–10 weeks of training with the Nordic hamstrings exercise (NHE) (Medeiros et al. in press; Mjolsnes et al., 2004), and a systematic review concluded that prevention programmes that include the NHE reduce by halving the risk of HSI among soccer players (Al Attar et al., 2017). Therefore, NHE seems to be a feasible and effective strategy to improve the H:Q torque ratios.

Although the predictive value of H:Q ratios has been criticised (Dauty et al., 2018; Van Dyk et al., 2017), prospective studies have found associations between H:Q ratios and HSI in soccer players (Croisier et al., 2002, 2008; Lee et al., 2018) and other athletes

(Orchard et al., 1997; Yeung et al., 2009). The lack of consensus may be explained, at least in part, by the complex multifactorial nature of sports injuries and the fact that they do not result from a linear interaction between isolated predictive factors, but rather from complex interactions between a 'web of determinants' (Bittencourt et al., 2016). However, further prospective studies support the hamstrings eccentric strength as a consistent HSI risk factor in athletes (Lee et al., 2018; Opar et al., 2015; Timmins et al., 2016). For instance, the univariate logistic regression performed by Timmins et al. (2016) demonstrated that the risk of HSI in soccer players was reduced by 9% for every 10 N increase in hamstrings eccentric strength. Together, those studies suggest that increasing the H:Q_{fun} ratio through the hamstrings strengthening hypothetically keep players away from the zone at greatest risk of HSI.

Like all sports injuries, HSI is multifactorial and most football players present multiple risk factors for sustaining this injury (Ribeiro-Alvares et al., 2019). Thus, a comprehensive approach is recommended for preventive programmes (Buckthorpe et al., 2019), while the present study was focused on only one aspect (i.e., muscle imbalance). Regarding this specific outcome, it is important to highlight that coaching staffs should always analyse the H:Q ratio concurrently with the PT values because a player with similar levels of weakness in quadriceps and hamstrings can be classified as 'normal' from the H:Q ratio data. Moreover, participants in the current study were tested in a single and slow angular velocity (i.e. 60°/s); but tests performed in higher angular velocities may provide further information (Stastny et al., 2018), given the nature of HSI during high-speed running. Lastly, the findings reported here should not be extrapolated to athletes engaged in other sports, since training-specific adaptations may produce strength profiles different than those found in soccer players.

In conclusion, this study demonstrated that hamstrings strength deficit is the key factor for low H:Q ratios in male soccer players, especially those with poor H:Q_{fun} ratio.

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Disclosure statement

No potential conflict of interest was reported by the author(s).

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